

Aligning Small LLMs with GRPO for Strict JSON Generation

Rule-Based Reward Optimization for Schema-Conformant Output

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Problem & Objective

- Small LLMs (<2B parameters) frequently fail to produce valid, schema-conformant JSON
- Common errors: unclosed brackets, type violations, conversational filler
- **Goal:** use GRPO with rule-based rewards to reach >90% Pass@1 on strict JSON
- No neural reward model → dense, **deterministic**, interpretable signal

Model	Params
SmolLM2-135M-Instruct	135M
SmolLM2-360M-Instruct	360M
Qwen2.5-0.5B-Instruct	0.5B
TinyLlama-1.1B-Chat	1.1B
Gemma-2-2B-it	2B

- 4-bit NF4 quantization
- LoRA $r=16$, $\alpha=32$
- 1× NVIDIA L40S (48 GB)
- 2 500 training steps / model

Group Relative Policy Optimization (GRPO)



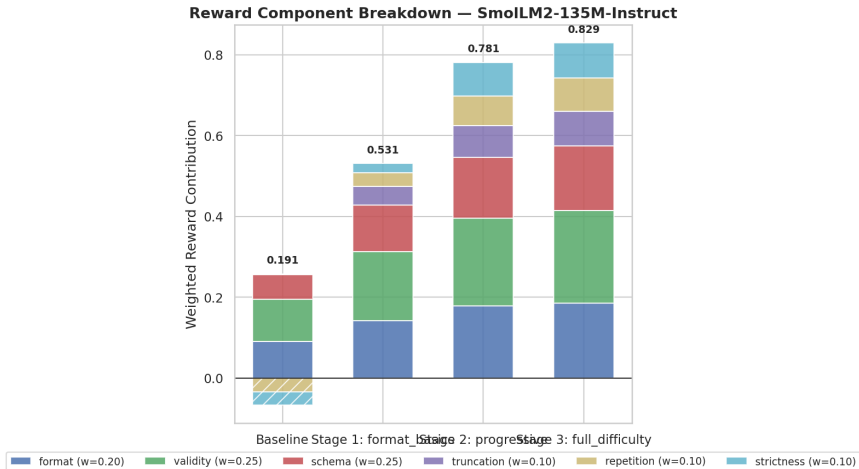
- No neural reward model — 7 **rule-based** components score each completion
- Additive rewards prevent **zero-advantage collapse**
- 4-bit NF4 + LoRA ($r=16$) on a single NVIDIA L40S GPU

Rule-Based Reward System

Component	Range	What it measures
Format	[0, 1]	“ ‘json“ ‘ code fence present
Validity	[0, 1]	JSON parseability (partial credit by error position)
Schema	[0, 1]	Structural conformance: keys, types, nesting depth
Reasoning	[−0.5, 1]	Quality of <think> block — Think mode only
Truncation	[−1, 1]	Unclosed brackets / trailing commas
Repetition	[−1, 1]	Degenerate loops, duplicate lines
Strictness	[−1, 1]	Extra text outside the JSON block

Additive score, not gating — gradient is always non-zero.

Reward Breakdown — SmolLM2-135M



No-Think / Curriculum — component contributions at final checkpoint

	Standard	Curriculum
No-Think	✓	✓
Think	✓	✓

Think: model generates <think> reasoning before JSON

Curriculum: 3 progressive stages

Stage	Steps	S / M / H	Temp
1 – Format Basics	800	35 / 55 / 10%	0.8
2 – Progressive	800	15 / 35 / 50%	0.7
3 – Full Difficulty	900	10 / 25 / 65%	0.6

S = Simple M = Medium H = Hard

Parametric Synthetic Dataset

- 24 parameterized templates
(8 per difficulty tier)
- 1500 samples/stage $\times$ 3 stages
= 4500 training samples
- Eval: **300 samples** — 100 simple / 100
medium / 100 hard
- **No ground-truth JSON**
reward functions evaluate directly

Difficulty tiers

Simple — flat key-value objects, typed arrays

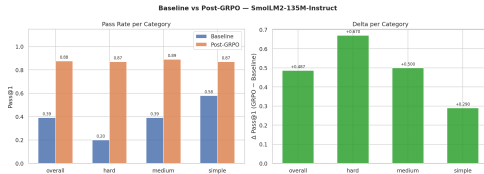
Medium — nested objects, config files, form
schemas

Hard — JSON Schema draft-07, paginated APIs,
OpenAPI specs

Results — Peak Performance (Pass@1)

Model	Base	Peak	Config	Δ
SmolLM2-135M	31%	90.0%	Think / Std	+58.7
SmolLM2-360M	78%	96.7%	NoThk / Std	+19.0
Qwen2.5-0.5B	92%	98.0%	Think / Curr	+6.3
TinyLlama-1.1B	79%	99.3%	NoThk / Curr	+20.0
Gemma-2-2B	97%	97.7%	Curriculum	+1.3

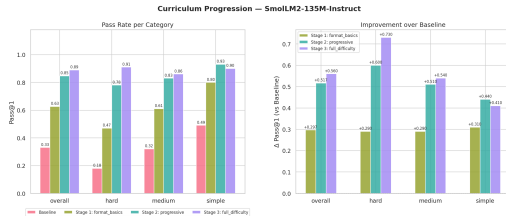
All models reach the 87–99% band.



SmolLM2-135M — baseline vs. GRPO (No-Think / Standard)

Standard vs. Curriculum Learning

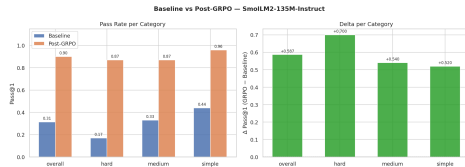
- **Standard** delivers strong gains for weak models
SmolLM2-135M: +48.67 pp, TinyLlama: +18.33 pp
- But causes **regression** on already-strong models
Gemma-2-2B: -1.67 pp on standard training
- **Curriculum's real value** is **preventing degradation**
Gemma-2-2B: standard -1.67 pp → curriculum +0.67 pp
- Marginal extra gain for weak models
135M: 87.67% → 89.00% (+1.33 pp only)



SmolLM2-135M — Pass@1 across curriculum stages

Think Mode — Reasoning Tokens in Practice

- **Think baseline is lower** for small models — the `<think>` prompt disrupts their fragile context
SmolLM2-360M: 77.67% → 69.33% at baseline alone
- Think vs. No-Think gain (standard training):
 - SmolLM2-135M **+2.33 pp**
 - SmolLM2-360M **-1.00 pp**
 - Qwen2.5-0.5B **+0.34 pp**
 - TinyLlama-1.1B **+1.00 pp**
 - Gemma-2-2B **-0.67 pp**
- Extra tokens → higher latency, negligible accuracy benefit



SmolLM2-135M Think / Standard — only model with a real Think-mode benefit

Notable anomaly — Gemma-2-2B Think/Curriculum

U-shaped curve: Stage 1 **83%** → Stage 2 **66%** → Stage 3 **97.67%**

Forcing reasoning on simple schemas temporarily collapses an already-aligned model

Lessons

- Additive rewards $>$ gating
(prevent zero-advantage collapse)
- Standard training is **sufficient** for weak models
- Curriculum is a **safety net** for strong models,
not a universal performance booster
- Think mode adds cost but **minimal accuracy gain**
for models $< 1\text{B}$ parameters
- 4-bit LoRA on a single GPU is sufficient

Limitations

- Evaluated on JSON only
(not directly generalizable)
- Synthetic dataset $\rightarrow$
potential real-world gap
- Single GPU, 2 500 steps only

Conclusions

- GRPO + rule-based rewards brings all 5 models to 87–99% Pass@1 on strict JSON generation
- **Standard training** is the go-to strategy for weak models — large gains with no curriculum overhead
- **Curriculum** is a safety net for strong models:
prevents the regression that standard training causes on Gemma-2-2B
- **Think mode** yields marginal gains not worth the latency cost — small models cannot generate meaningful reasoning tokens